

Calculus I

Review of basic functions

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2019

Outline

1 A Catalog of Essential Functions

- Linear Functions
- Polynomials
- Power Functions
- Rational Functions
- Algebraic Functions
- Transcendental Functions

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 - Linear Functions
 - Polynomials
 - Power Functions
 - Rational Functions
 - Algebraic Functions
 - Transcendental Functions
- 2 New Functions from Old Functions

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Linear Functions

Definition (Linear Function)

A linear function is a function the graph of which is a line. We can write any linear function in slope-intercept form:

$$f(x) = mx + b.$$

m is called the slope, and b is called the y -intercept.

$f(x)$	Direction	y -intercept
$x + 1$ $-0.5x$ -1		

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- $m > 0$ means the graph of f points up ().

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- $m > 0$ means the graph of f points up ($\nearrow$).
- $m < 0$ means the graph of f points down ($\searrow$).

$f(x)$	Direction	y -intercept
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- $m > 0$ means the graph of f points up ().
- $m < 0$ means the graph of f points down ().
- $m = 0$ means the graph of f is horizontal (.

$f(x)$	Direction	y -intercept
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-1		

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- $m = 0$ means the graph of f is horizontal ().
- b tells us the height of the point where the graph hits the y -axis.

$f(x)$	Direction	y -intercept
$x + 1$		1
$-0.5x + 0$		0
-1		

- $m > 0$ means the graph of f points up ().
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- $m = 0$ means the graph of f is horizontal ().
- b tells us the height of the point where the graph hits the y -axis.

Polynomials

Definition (Polynomial Function)

A polynomial function is a function f of the form

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1} + a_nx^n,$$

where n is a non-negative integer and $a_0, \dots, a_n$ are real numbers, called the coefficients. If $a_n \neq 0$ the integer n is called the degree of f .

If we interpret x as an indeterminate formal expression, rather than a number, we say that $f(x)$ is a polynomial (rather than a polynomial function).

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$f(x)$	Polynomial?	Degree	a_0	a_1	a_2
$x^4 - x + 1$					
6					
$3x^2 - \frac{1}{2}x + \sqrt{x}$					
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$f(x)$	Polynomial?	Degree	a_0	a_1	a_2
$x^4 - x + 1$	Yes	4			
6					
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$x^4 - x + 1$	Yes	4	1	-1	0
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Polynomials

Definition (Polynomial Function)

A polynomial function is a function f of the form

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1} + a_nx^n,$$

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$f(x) = x^{-1} = \frac{1}{x}$ is called the reciprocal function. Its graph has equation $y = \frac{1}{x}$, or $xy = 1$, and is an hyperbola with the coordinate axes as its asymptotes.

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Example ($x/(x^2 - 1)$)

The function

$$f(x) = \frac{x}{x^2 - 1}$$

is a rational function.

Algebraic Functions

(Algebraic Function)

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Outside of present course: function $f(x)$ = algebraic if it satisfies a polynomial equation with polynomial coefficients, i.e., $a_0(x) + a_1(x)f(x) + \cdots + a_n(x)(f(x))^n = 0$ for some polynomials $a_i(x)$.

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Examples.

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- And many more.
- Outside of the present course: by definition, a function is transcendental if it is not algebraic, i.e., if it satisfies no polynomial equation with polynomial coefficients.

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Definition (Composition of f and g)

If f and g are two functions, then the composition of f and g is written $f \circ g$ and is defined by the formula

$$(f \circ g)(x) = f(g(x)).$$

Imagine f and g as machines taking some input and producing some output. Then $f \circ g$ corresponds to attaching both machines end-to-end so that the output of g becomes the input of f .

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The domain of $f \circ g$ is the set of all numbers x in the domain of g such that $g(x)$ is in the domain of f . If the domain of f is A and the domain of g is B , we write this as

$$\{x \in B \mid g(x) \in A\}.$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

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$$3-x \geq 0$$

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$$3-\sqrt{x} \geq 0$$

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Domain :

$$x \geq 0$$

$$3 - \sqrt{x} \geq 0$$

$$-\sqrt{x} \geq -3$$

Example

Find $f \circ g, g \circ f, g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{3-x}) = \sqrt{\sqrt{3-x}} = \sqrt[4]{3-x}$$

Domain:

$$3-x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$x \in (-\infty, 3].$$

$$(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{3-\sqrt{x}}$$

Domain :

$$x \geq 0$$

$$3-\sqrt{x} \geq 0$$

$$-\sqrt{x} \geq -3$$

$$\sqrt{x} \leq 3$$

Example

Find $f \circ g, g \circ f, g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{3-x}) = \sqrt{\sqrt{3-x}} = \sqrt[4]{3-x}$$

Domain:

$$3-x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$x \in (-\infty, 3].$$

$$(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{3-\sqrt{x}}$$

Domain :

$$x \geq 0$$

$$3-\sqrt{x} \geq 0$$

$$-\sqrt{x} \geq -3$$

$$\sqrt{x} \leq 3$$

$$x \leq 9$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{3-x}) = \sqrt{\sqrt{3-x}} = \sqrt[4]{3-x}$$

Domain:

$$3-x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$x \in (-\infty, 3].$$

$$(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{3-\sqrt{x}}$$

Domain :

$$x \geq 0$$

$$3-\sqrt{x} \geq 0$$

$$-\sqrt{x} \geq -3$$

$$\sqrt{x} \leq 3$$

$$x \leq 9$$

$$x \in ?$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{3-x}) = \sqrt{\sqrt{3-x}} = \sqrt[4]{3-x}$$

Domain:

$$3-x \geq 0$$

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$$x \leq 3$$

$$x \in (-\infty, 3].$$

$$(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{3-\sqrt{x}}$$

Domain :

$$x \geq 0$$

$$3-\sqrt{x} \geq 0$$

$$-\sqrt{x} \geq -3$$

$$\sqrt{x} \leq 3$$

$$x \leq 9$$

$$x \in [0, 9]$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x)$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x))$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x})$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$3 - \sqrt{3-x} \geq 0$$

Example

Find $f \circ g$, $g \circ f$, $g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$3 - \sqrt{3-x} \geq 0$$

$$-\sqrt{3-x} \geq -3$$

Example

Find $f \circ g, g \circ f, g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$3 - \sqrt{3-x} \geq 0$$

$$-\sqrt{3-x} \geq -3$$

$$\sqrt{3-x} \leq 3$$

Example

Find $f \circ g, g \circ f, g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$3 - \sqrt{3-x} \geq 0$$

$$-\sqrt{3-x} \geq -3$$

$$\sqrt{3-x} \leq 3$$

$$3 - x \leq 9$$

Example

Find $f \circ g, g \circ f, g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$3 - \sqrt{3-x} \geq 0$$

$$-\sqrt{3-x} \geq -3$$

$$\sqrt{3-x} \leq 3$$

$$3 - x \leq 9$$

$$-x \leq 6$$

Example

Find $f \circ g, g \circ f, g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$3 - \sqrt{3-x} \geq 0$$

$$-\sqrt{3-x} \geq -3$$

$$\sqrt{3-x} \leq 3$$

$$3 - x \leq 9$$

$$-x \leq 6$$

$$x \geq -6$$

Example

Find $f \circ g, g \circ f, g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$3 - \sqrt{3-x} \geq 0$$

$$-\sqrt{3-x} \geq -3$$

$$\sqrt{3-x} \leq 3$$

$$3 - x \leq 9$$

$$-x \leq 6$$

$$x \geq -6$$

$$x \in ?$$

Example

Find $f \circ g, g \circ f, g \circ g$ and their domains, where $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$.

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{3-x}) = \sqrt{3 - \sqrt{3-x}}$$

Domain :

$$3 - x \geq 0$$

$$-x \geq -3$$

$$x \leq 3$$

$$3 - \sqrt{3-x} \geq 0$$

$$-\sqrt{3-x} \geq -3$$

$$\sqrt{3-x} \leq 3$$

$$3 - x \leq 9$$

$$-x \leq 6$$

$$x \geq -6$$

$$x \in [-6, 3].$$

Example

Give simplified formulas for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x - 1}{x + 2}$$

$$g(x) = \frac{2x + 3}{5x - 7}$$

Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x - 1}{x + 2}$$

$$g(x) = \frac{2x + 3}{5x - 7}$$

$x \neq ?$

Example

Give simplified formulas for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$\begin{array}{l} f(x) = \frac{2x - 1}{x + 2} \\ g(x) = \frac{2x + 3}{5x - 7} \end{array} \quad \left| \quad x \neq -2 \right.$$

Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x - 1}{x + 2}$$

$$g(x) = \frac{2x + 3}{5x - 7}$$

$$x \neq -2$$

$$x \neq ?$$

Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x - 1}{x + 2}$$

$$g(x) = \frac{2x + 3}{5x - 7}$$

$$x \neq -2$$

$$x \neq \frac{7}{5}$$

Example

Give simplified formulas for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x - 1}{x + 2}$$

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$$x \neq -2$$

$$x \neq \frac{7}{5}$$

$$(f \circ g)(x) = f(g(x))$$

Example

Give simplified formulas for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x - 1}{x + 2}$$

$$g(x) = \frac{2x + 3}{5x - 7}$$

$$x \neq -2$$

$$x \neq \frac{7}{5}$$

$$(f \circ g)(x) = f(g(x)) = f\left(\frac{2x + 3}{5x - 7}\right)$$

Example

Give simplified formulas for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x - 1}{x + 2}$$

$$g(x) = \frac{2x + 3}{5x - 7}$$

$$x \neq -2$$

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$$(f \circ g)(x) = f(g(x)) = f\left(\frac{2x + 3}{5x - 7}\right) = \frac{2\left(\frac{2x + 3}{5x - 7}\right) - 1}{\frac{2x + 3}{5x - 7} + 2}$$

Example

Give simplified formulas for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x - 1}{x + 2}$$

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Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

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Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x-1}{x+2}$$

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$$\begin{aligned} (f \circ g)(x) &= f(g(x)) = f\left(\frac{2x+3}{5x-7}\right) = \frac{2\left(\frac{2x+3}{5x-7}\right) - 1}{\frac{2x+3}{5x-7} + 2} \\ &= \frac{\frac{2(2x+3)}{5x-7} - \frac{5x-7}{5x-7}}{\frac{2x+3}{5x-7} + \frac{2(5x-7)}{5x-7}} = \frac{\frac{4x+6-(5x-7)}{5x-7}}{\frac{2x+3+(10x-14)}{5x-7}} \end{aligned}$$

Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

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Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

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Example

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Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

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Example

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Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x-1}{x+2}$$

$$x \neq -2$$

$$g(x) = \frac{2x+3}{5x-7}$$

$$x \neq \frac{7}{5}$$

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) = f\left(\frac{2x+3}{5x-7}\right) = \frac{2\left(\frac{2x+3}{5x-7}\right) - 1}{\frac{2x+3}{5x-7} + 2} \\ &= \frac{\frac{2(2x+3)}{5x-7} - \frac{5x-7}{5x-7}}{\frac{2x+3}{5x-7} + \frac{2(5x-7)}{5x-7}} = \frac{\frac{4x+6-(5x-7)}{5x-7}}{\frac{2x+3+(10x-14)}{5x-7}} = \frac{-x+13}{12x-11} \end{aligned}$$

$$x \neq ?$$

Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

$$f(x) = \frac{2x-1}{x+2}$$

$$x \neq -2$$

$$g(x) = \frac{2x+3}{5x-7}$$

$$x \neq \frac{7}{5}$$

$$(f \circ g)(x) = f(g(x)) = f\left(\frac{2x+3}{5x-7}\right) = \frac{2\left(\frac{2x+3}{5x-7}\right) - 1}{\frac{2x+3}{5x-7} + 2}$$

$$= \frac{\frac{2(2x+3)}{5x-7} - \frac{5x-7}{5x-7}}{\frac{2x+3}{5x-7} + \frac{2(5x-7)}{5x-7}} = \frac{\frac{4x+6-(5x-7)}{5x-7}}{\frac{2x+3+(10x-14)}{5x-7}} = \frac{-x+13}{12x-11}$$

$$x \neq \frac{11}{12}, \frac{7}{5}$$

Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

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$$x \neq -2$$

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Example

Give simplified f-las for $f \circ g$, $f \circ f$, $g \circ f$, $g \circ g$. Find the implied domains.

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$$(f \circ f)(x) = f(f(x))$$

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$$(g \circ f)(x) = ?$$

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$$x \neq -2, -\frac{3}{4}$$

$$(g \circ f)(x) = \frac{7x+4}{3x-19}$$

$$x \neq -2, \frac{19}{3}$$

$$(g \circ g)(x) = \frac{19x-15}{-25x+64}$$

$$x \neq \frac{7}{5}, \frac{64}{25}$$